

Auditing Closed-Loop Learning in Recurrent Neural Networks: Reproduction, Robustness, and Generalization

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Paper under double-blind review

Abstract

Recurrent neural networks are often used as mechanistic models of learning and control, but closed-loop training creates reproducibility challenges because a model’s actions alter future inputs. We conduct a claim-level reproducibility study of Ger and Barak’s closed-loop RNN learning dynamics, testing independent implementation, seed variation, protocol perturbations, coupled-system diagnostics, and architecture/task transfer. The central closed/open divergence reproduces strongly in the original double-integrator setting: 50/50 paired seeds favor closed-loop training, with mean final deployed losses 0.0933 versus 0.3119. The stage claim is more conditional: all original seeds show early rapid improvement followed by slow progress, but objective three-stage changepoint detectors do not recover a robust second boundary. A targeted A1 analysis separates stability and behavioral trade-offs: short-horizon improvements coincide with coupled-radius increases in 120/120 runs, while the loss-only short-vs-long criterion is supported in 0/120. Generalization is hierarchical: architecture transfer within the double-integrator-style task is strong, tracking transfer is partial, and ring path-integration transfer is weak. These results motivate reporting practices for closed-loop RNN studies: deployed closed-loop loss, paired seeds, algorithmic stage criteria, coupled-system spectra, feedback strength, rollout horizon, and failure rates.

1 Introduction

Closed-loop recurrent neural networks occupy an important middle ground between supervised sequence modeling, control, and computational neuroscience. In a closed-loop environment, an RNN’s output changes the future inputs it receives. This feedback makes the training problem qualitatively different from open-loop imitation or teacher-forced supervised learning, where future inputs are independent of the learner’s deployed actions. Ger and Barak (Ger and Barak, 2025) recently argued that this distinction produces learning dynamics that are fundamentally different from those of otherwise identical open-loop RNNs. Their paper is valuable because it makes a mechanistic claim rather than only a performance claim: the coupled agent-environment system, not the RNN alone, determines important learning transitions.

This paper is not a figure-level reproduction study. Instead, we use Ger and Barak’s closed-loop RNN learning framework as a test case for claim-level reproducibility in recurrent control systems. We ask which conclusions survive independent implementation, seed variation, optimizer and hyperparameter perturbations, coupled-system stability analysis, and transfer to nonlinear or path-integration-style settings. This framing matters for TMLR-style reproducibility: a rerun of a public artifact can be educational, but it is only broadly useful when it surfaces reusable lessons about when claims are reliable, when they are protocol-sensitive, and what future papers should report.

Our study makes four contributions. First, we provide an independent, config-driven reproduction of the central open-loop/closed-loop learning comparisons in Ger and Barak. Second, we convert qualitative claims about learning stages and stability transitions into quantitative, seed-level tests with confidence intervals. Third, we measure robustness under perturbations to optimizer, initialization, feedback strength, noise,

and rollout horizon, identifying which conclusions are stable and which are protocol-sensitive. Fourth, we distill an audit checklist for future closed-loop RNN studies, including paired-seed comparisons, deployed closed-loop loss, stage criteria, coupled-system stability diagnostics, and failure-rate reporting.

2 Related Work

Task-trained RNNs in computational neuroscience. Task-trained RNNs are widely used as mechanistic models because they can solve cognitive and motor tasks while exposing internal dynamics for analysis. Early and influential work showed that high-dimensional trained RNNs can implement low-dimensional dynamical mechanisms (Sussillo and Barak, 2013), and that recurrent dynamics can account for context-dependent computation in prefrontal cortex (Mante et al., 2013). This line of work has become a general modeling strategy in systems neuroscience, where RNNs serve as hypothesis-generating models rather than only predictive tools (Barak, 2017). At the same time, large-scale analyses of trained recurrent networks show that architecture and training details can change representational geometry and dynamics even when task performance is similar (Maheswaranathan et al., 2019). This motivates our claim-level framing: reproducing a task curve is not enough if the mechanistic conclusion depends on training protocol or model class.

Dynamical-systems analysis of RNNs. The original paper sits in a tradition of reverse-engineering trained RNNs through fixed points, low-dimensional projections, eigenvalues, and linearized dynamics. Sussillo and Barak’s low-dimensional analysis made the case that trained recurrent networks can be understood through dynamical-systems tools (Sussillo and Barak, 2013), and later work used related analyses to compare recurrent dynamics across large populations of networks (Maheswaranathan et al., 2019). Our audit differs in emphasis. We do not only ask whether a trained RNN has interpretable internal dynamics; we ask whether the relevant dynamical object is the RNN alone or the coupled agent-environment system. This distinction is central in closed-loop settings because the environment transforms actions into future observations.

Open-loop training, closed-loop deployment, and distribution shift. The mismatch between teacher-forced training and deployed rollouts is well known in sequence prediction and imitation learning. Scheduled sampling was proposed to reduce the discrepancy between training on ground-truth previous tokens and testing on model-generated tokens (Bengio et al., 2015). In imitation learning, DAGger and related reductions explicitly address the fact that a learner’s actions change the future states it visits, invalidating i.i.d. supervised-learning assumptions (Ross et al., 2011). Closed-loop RNN learning has an analogous but mechanistically richer mismatch: open-loop imitation can minimize teacher-forced action error while still failing under deployed closed-loop rollouts. This connection motivates our insistence on reporting deployed closed-loop loss, not only open-loop or imitation loss.

Reproducibility and robustness in machine learning. Our audit follows the broader shift from artifact reruns to structured reproducibility studies. The NeurIPS reproducibility program emphasized clear artifacts, documented environments, and reproducibility checklists (Pineau et al., 2021). In deep reinforcement learning, Henderson et al. showed that algorithmic conclusions can be sensitive to seeds, evaluation protocol, and implementation details (Henderson et al., 2018). Closed-loop RNN studies share many of these risks because training outcomes can depend on random initialization, rollout horizon, optimizer, and environment coupling. The present paper therefore treats Ger and Barak’s work as a case study for a reusable audit protocol: paired seeds, stress tests, coupled-system diagnostics, and reporting lessons.

We use *reproduction* for tests that preserve the original task family and claim, *robustness* for perturbations of seeds, optimizer, initialization, horizon, noise, feedback, and clipping, and *generalization* for architecture or task-family changes. This terminology is intentionally conservative: a robustness or generalization failure does not by itself falsify the original result, but it does delimit where the claim should be reported as reliable.

3 Original Claims and Audit Questions

We separate the original claims under reproduction from audit questions introduced by this study. The reproduced claims are C1–C3. C1 is the central divergence claim: identical RNNs trained open-loop and

Table 1: Direct mapping from original evidence to our reproduction tests. This separates figure-level reproduction from claim-level decisions.

Original claim	Original evidence	Our test	Result	Decision
Closed/open divergence	Main loss and gain figures	50 paired seeds, shared initialization, deployed loss	50/50 seeds; mean loss gap 0.2186	Reproduced
Stage structure	Visual learning-stage plots	Derivative detector plus change-points	slow phase 50/50; strict three-stage 0/50	Partial
Coupled stability	Coupled eigenvalue argument	Coupled vs RNN-only spectra	50/50 coupled stability crossings	Reproduced, restricted
Motor-control applicability	Tracking extension	Tracking plus ring variants	tracking 8/10; ring-family 4/50	Partial boundary

closed-loop follow different learning trajectories and differ under closed-loop deployment. C2 is the stage claim: closed-loop learning passes through distinct learning stages. C3 is the mechanistic stability claim: progress depends on stability of the coupled agent-environment system.

The audit questions are A1–A2. A1 asks whether C1–C3 are robust to protocol perturbations and whether a targeted short-vs-long horizon analysis reveals the expected stability tradeoff. A2 asks whether the phenomena generalize across architectures and tasks. We use A1 and A2 to avoid implying that Ger and Barak made every robustness or generalization claim in exactly our framing.

For each reproduced claim or audit question, we define three evidence levels. A result is *reproduced* when the effect and diagnostic hold under the relevant protocol for most paired seeds, with confidence intervals excluding zero or a pre-specified smallest effect size of interest. For C1 this smallest effect is a 5 percent relative deployed-loss gap. A result is *partially reproduced* when the direction appears but is unstable across seeds, sensitive to plausible perturbations, or only weakly supported by the diagnostic. A result is *not reproduced* when the effect is absent, reverses, or depends on hand-picked visual criteria.

4 Closed-Loop RNN Reproducibility Audit Protocol

We specify a modest audit protocol rather than proposing a new benchmark. The protocol is a recipe for testing claims about closed-loop RNN learning dynamics in any recurrent control setting where actions can affect future observations. Its inputs are: a set of original claims, a task/environment, an open-loop training procedure, a closed-loop training procedure, a seed set, a small set of protocol perturbations, and at least one held-out architecture or task variant. Its outputs are: per-seed deployed closed-loop metrics, stage labels, stability diagnostics, perturbation summaries, generalization summaries, and a claim-level reproducibility matrix. Table 2 gives the concrete specification used in this paper.

The protocol deliberately separates *measurement* from *interpretation*. Measurement consists of paired runs, deployed losses, spectra, stage labels, and perturbation outcomes. Interpretation happens only after aggregating seed-level outputs into claim-level decisions. This separation is important because closed-loop RNN papers often make qualitative statements about stages or mechanisms from a small number of curves; our protocol requires those statements to be tied to explicit detectors, uncertainty estimates, and failure rates.

4.1 Metrics and run accounting

The main metrics are deployed closed-loop loss, teacher-forced imitation loss, effective feedback gain, coupled and RNN-only spectral radius, algorithmic stage labels, and A1 multi-horizon tradeoff components. Exact formulas are in Appendix A. The 455 completed paired runs consist of 50 core C1–C3 runs, 195 robustness runs, 120 targeted A1 multi-horizon runs, and 90 A2 extension runs. Detector thresholds were selected during smoke-test development before running the full reported sweeps; threshold sensitivity is reported to show which conclusions depend on those choices.

Table 2: Closed-loop RNN reproducibility audit protocol. The protocol is intentionally lightweight: each row defines a required audit step, its minimum implementation, and the output needed to support claim-level conclusions.

Audit step		Minimum requirement	Recorded output
Paired training		Clone the same initialization into open-loop and closed-loop learners; keep architecture and evaluation budget fixed.	Per-seed train loss, deployed closed-loop test loss, peak deployed loss, final loss, and effective-gain trajectory for both learners.
Seed-level	uncertainty	Run enough seeds to estimate direction agreement and confidence intervals; report seed count explicitly.	Mean, 95 percent bootstrap CI, fraction of seeds supporting each claim, and seed-level failure/recovery rates.
Stage detection		Predefine a detector using log-loss derivative, minimum plateau duration, and optional stability crossing.	Stage labels, plateau duration, plateau loss, transition epochs, and frequency of detected stages across seeds/settings.
Coupled diagnostics		Compute RNN-only and coupled agent-environment spectra whenever the system can be linearized or locally linearized.	Spectral radius time series, stability-crossing epoch, plateau-exit gap, and predictive correlation with recovery/failure.
Robustness perturbations		Change factors that alter optimization or environment coupling, such as optimizer, initialization scale, feedback strength, noise, horizon, control penalty, and clipping.	Claim support under each perturbation, effect-size changes, and perturbation-level failure rates.
Generalization check		Test at least one architecture or task variant that preserves action-dependent inputs and one that changes the recurrent/control structure.	Claim support by variant and a boundary-condition statement explaining when the phenomenon does or does not transfer.
Claim/audit	decision	Classify each claim or audit question as reproduced, partially reproduced, or not reproduced using the same criteria across settings.	Matrix linking evidence type, original setting, robustness, generalization, and actionable lesson.

4.2 Paired open-loop and closed-loop training

For each seed, one initialization is cloned into an open-loop and a closed-loop learner. The closed-loop model is trained by rolling out in the environment; the open-loop model imitates teacher trajectories. Both are evaluated under deployed closed-loop rollout.

4.3 Stage detection

The original paper describes closed-loop learning stages visually. We operationalize C2 as early rapid improvement, a plateau or slow-progress phase, and post-transition improvement, detected from log-loss derivatives, a minimum plateau duration, and optionally a coupled spectral-radius crossing.

4.4 Coupled-system stability diagnostics

For linearized settings, we compute the spectral radius of the RNN recurrent matrix and of the coupled agent-environment transition matrix. This targets C3 directly: if closed-loop dynamics are governed by the agent-environment loop, the coupled diagnostic should be more informative than an isolated RNN diagnostic.

4.5 Robustness and generalization

The A1 robustness sweep changes learning rate, initialization scale, feedback strength, episode length, observation noise, control penalty, optimizer, and gradient clipping. The A2 generalization sweep tests nonlinear RNNs, GRUs, low-rank RNNs, tracking control, and ring/path-integration-style tasks.

4.6 Original artifact, independent implementation, and deviations

We preserved the public artifact under `external/original_artifact/`. It contains the original notebooks for figure reproduction, nonlinear training, theory, and tracking, plus the pickle files used by the artifact. Our reproduction log records which files load, which notebooks depend on precomputed data, and which notebook runs are long or checkpoint-heavy.

The experiments reported here use an independent, config-driven implementation rather than direct notebook execution. The double-integrator task, paired open/closed training, effective gains, coupled spectra, stage detectors, robustness perturbations, and generalization tasks were reimplemented as package modules and command-line scripts. We matched the original double-integrator architecture and training profile where needed for C1–C3, then intentionally introduced new A1/A2 audits over seeds, optimizer, initialization, feedback, horizon, noise, clipping, architecture, and task. We did not contact the original authors; all choices were made from the paper, public artifact, and documented configs. Exact notebook reruns are therefore treated as artifact checks, while the paper’s quantitative claims are evaluated through the independent implementation. Every table and figure can be regenerated from saved raw CSVs with `scripts/run_full_audit.sh`, `scripts/run_targeted_c2_a1_a2.sh`, and `python -m closed_loop_repro.plotting.make_all_figures -config configs/figures/tmlr.yaml`.

A code-level comparison shows that the core reproduction matches the original double-integrator dynamics, hidden size, tanh controller class, SGD learning rate, batch size, rollout horizon, gradient clipping, and teacher-student open-loop design. It is not a line-for-line reimplement: the original nonlinear notebook uses a zero first action, current-state loss accumulation, no control penalty in that script, 1001 epochs, and smaller input/output initializations under its small-scale setting. We therefore interpret the package as a faithful claim-level reproduction of the original protocol family, not an exact numerical rerun of every notebook.

5 Result 1: Direct Reproduction of the Core Divergence

The central open-loop/closed-loop divergence reproduces cleanly under the original double-integrator setting (Figure 1). Across 50 paired seeds, the final deployed closed-loop loss of the closed-loop trained model is 0.0933 with bootstrap 95 percent CI [0.0932, 0.0935], seed SD 0.00054, and IQR [0.0929, 0.0937]. The open-loop trained student, evaluated under deployed closed-loop rollout, has final loss 0.3119 with CI [0.3044, 0.3193], seed SD 0.0269, and IQR [0.2919, 0.3255]. The paired deployed-loss gap is therefore 0.2186 with CI [0.2110, 0.2260], seed SD 0.0268, and direction agreement in 50/50 seeds. Effective feedback gains also diverge: the final closed/open gain distance is 0.5045 with CI [0.4998, 0.5092].

The result is not just a visual overlay of curves. It shows that minimizing open-loop imitation loss can produce a controller whose deployed closed-loop loss remains substantially higher than the directly closed-loop trained controller. This supports C1 and the main reporting lesson of the audit: closed-loop RNN studies should report deployed closed-loop performance, not only teacher-forced or open-loop training loss.

6 Result 2: Robustness and Tradeoff Audit

The robustness sweep consists of 195 paired runs: 13 perturbation settings with 15 seeds per setting (Table 3 and Figure 3). C1 is robust across most perturbations. It is supported in 165/195 robustness runs overall and in every finite run for the original, learning-rate, initialization, feedback-strength, episode-length, observation-noise, and high-control-penalty perturbations. The two important exceptions are informative. With Adam, the final deployed loss gap almost disappears: mean final closed-loop loss is 0.0780 and mean final open-loop deployed loss is 0.0798, yielding no C1 support under the 5 percent deployed-loss criterion. With gradient clipping disabled, all 15 runs in the `no_clip` condition produce non-finite final losses and are counted as failures rather than negative evidence for the mechanistic claim.

The stage claim is more sensitive to how “stage” is defined (Figure 2). A slow-progress phase is detected in 180/195 robustness runs, with failures concentrated entirely in the non-finite no-clipping condition. However, the stricter three-stage criterion, which requires both a plateau and a non-fallback second boundary, is

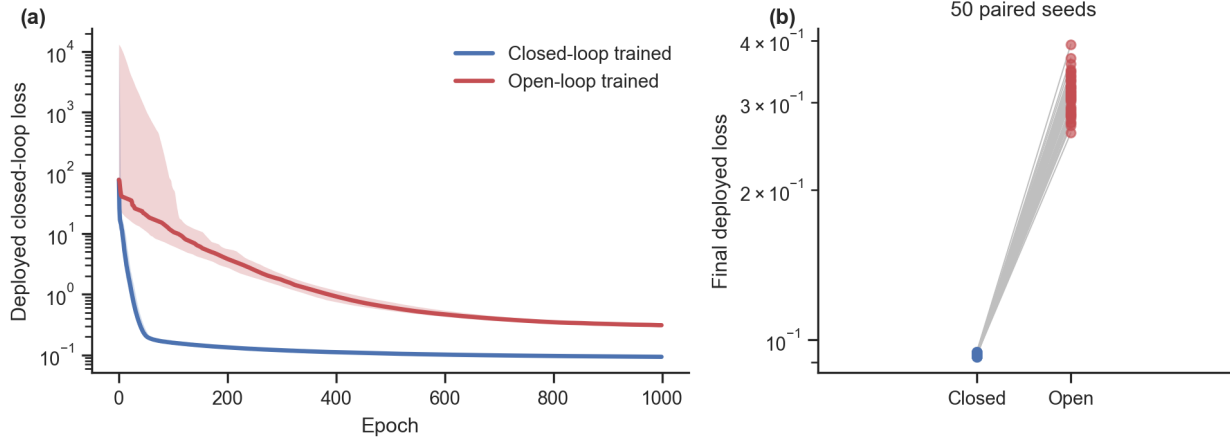


Figure 1: Core C1 reproduction under the original double-integrator setting. (a) Median deployed closed-loop test loss across paired seeds, with interquartile bands. (b) Paired final deployed losses for the same initializations. The direction of the deployed-loss gap reproduces in 50/50 seeds.

Table 3: Robustness perturbation grid. The original protocol also has the separate 50-seed core reproduction; the robustness grid repeats the baseline for 15 additional paired seeds so every perturbation has the same seed count.

Perturbation	Changed value	Seeds
original	Baseline double-integrator protocol	50 core; 15 grid
lr_low	learning rate 0.003	15
lr_high	learning rate 0.03	15
init_weak	recurrent initialization scale 0.03	15
init_strong	recurrent initialization scale 0.3	15
feedback_low	environment feedback strength 0.5	15
feedback_high	environment feedback strength 1.5	15
episode_short	rollout horizon 25 steps	15
episode_long	rollout horizon 100 steps	15
obs_noise	observation noise standard deviation 0.05	15
adam	optimizer Adam with learning rate 0.001	15
high_control_penalty	control penalty 0.05	15
no_clip	gradient clipping disabled	15

supported in only 1/195 robustness runs. Thus the broad qualitative observation that closed-loop training can enter a slow-progress regime is robust, but the claim that three distinct stage boundaries are algorithmically stable is not supported by the current detector.

The coupled-system stability transition is robust wherever the coupled spectrum is available and the run remains finite. It is supported in 180/195 robustness runs, again failing only in the no-clipping condition. This supports a restricted version of the original mechanistic emphasis in linearizable settings and separates the stability claim from the more fragile stage-boundary claim.

The targeted A1 sweep is more specific than the original union metric suggests (Figure 4). Across six control penalties and 20 seeds per penalty, the union criterion—short-horizon improvement plus either long-horizon loss worsening or increased coupled spectral radius—is supported in 120/120 runs, with mean conditional fraction 0.2814. Splitting the criterion shows that the effect is carried almost entirely by the spectral term: short-horizon improvement plus increased coupled radius is supported in 120/120 runs, whereas the loss-only criterion is supported in 0/120 runs and has mean conditional fraction 0.0038. This distinction prevents the

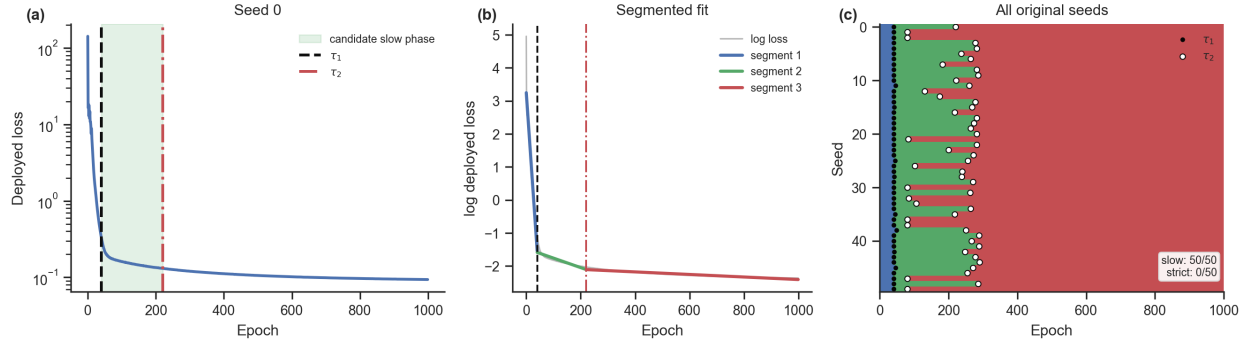


Figure 2: Stage analysis for the original-setting runs. (a,b) A representative seed with candidate boundaries τ_1 and τ_2 and the shaded candidate slow-progress phase. (c) Aggregate stage raster for all 50 original seeds; black dots mark τ_1 and white dots mark τ_2 . The segmented detector uses minimum segment length 20 epochs, Stage 1 slope < -0.02 , Stage 2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage 3 reacceleration slope $< \text{slope}_2 - 0.002$. It finds rapid-improvement plus slow-progress evidence in 50/50 seeds but strict three-stage support in 0/50.

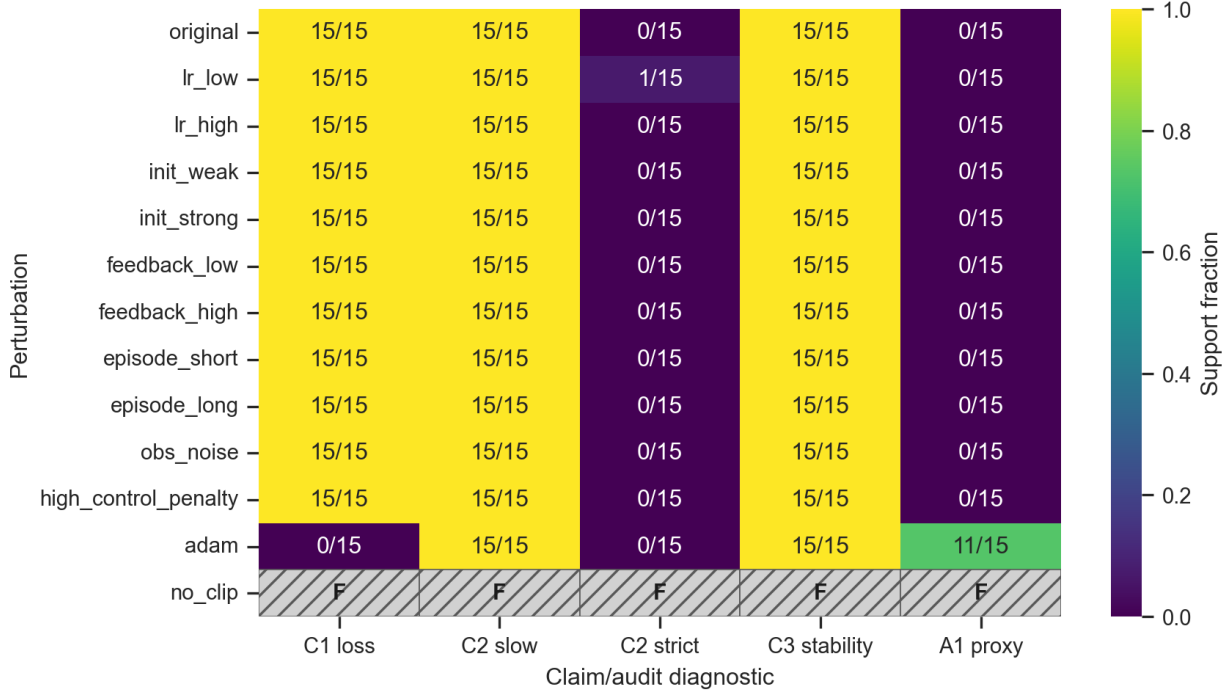


Figure 3: Robustness heatmap over perturbation settings and claim diagnostics. C1 and C3 are stable across most finite settings, C2 is robust only under the broad slow-progress criterion, the weak A1 proxy is not decisive outside the targeted multi-horizon sweep, and disabling gradient clipping produces non-finite failures marked as F rather than zero support.

audit from overstating a behavioral loss tradeoff: the detected A1 signal is a stability-cost signal, not a deployed-loss reversal.

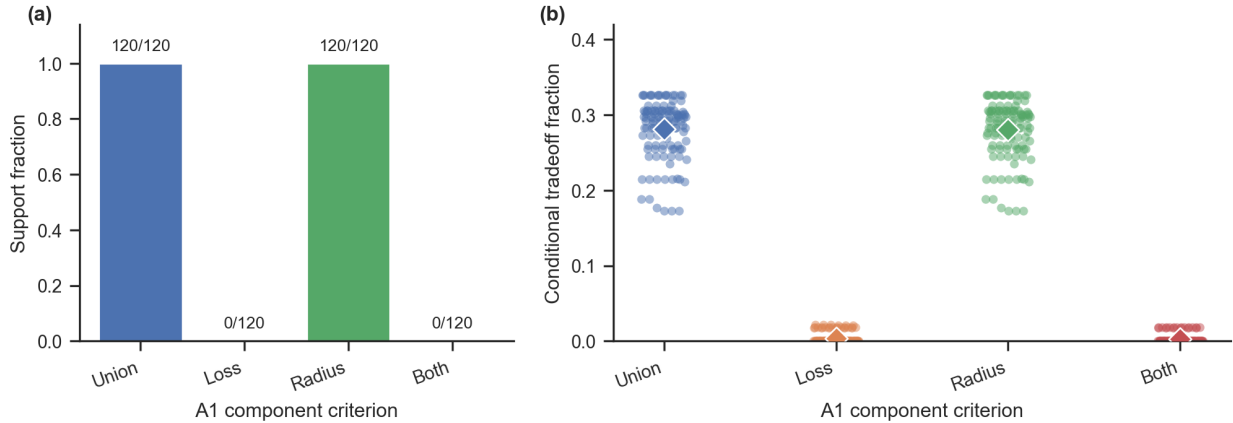


Figure 4: Targeted A1 component analysis over 120 paired tradeoff runs. (a) Support counts for the union criterion, loss-only criterion, radius criterion, and the intersection of loss and radius criteria. (b) Seed-level conditional fractions among short-horizon improvement intervals. The broad union result is 120/120, but the loss-only short-vs-long criterion is 0/120; the signal is dominated by coupled-radius increases. Diamonds show means.

Table 4: C2 threshold sensitivity on the 50 original-setting seeds. Slow-progress support is stable, but strict three-stage support remains absent across looser, stricter, and shorter-segment changepoint variants.

Detector variant	Slow phase	Strict three-stage	Median τ_1, τ_2
Main segmented	50/50	0/50	40, 255
Loose segmented	50/50	0/50	40, 255
Strict segmented	47/50	0/50	40, 255
Short-segment binary style	49/50	0/50	20, 70
Very loose reacceleration	50/50	0/50	40, 255

7 Result 3: Which Diagnostics are Predictive?

The original-setting coupled-system stability transition reproduces in 50/50 seeds (Figure 5). The final coupled spectral radius is finite and below the stability threshold in every original run, and the stability-crossing epoch occurs at the same point identified by the derivative-based plateau-start detector. This supports a restricted version of C3 in linearizable settings: coupled-system spectra recover the stability transition more directly than RNN-only spectra.

However, the spectral diagnostic is architecture-dependent in the current implementation. For linear, tanh, and low-rank controllers on linear tasks, the coupled transition matrix can be constructed explicitly. We do not report coupled spectral diagnostics for GRU or ring variants because our predefined C3 diagnostic requires an explicitly constructed coupled transition matrix. Extending C3 to gated nonlinear controllers requires local Jacobian linearization, which we leave to future work.

The most important diagnostic distinction concerns C2. The coupled stability crossing is robust, but the strict three-stage plateau-exit boundary is not. In the original setting, a slow-progress phase is present in 50/50 seeds, yet the second boundary is a fallback in 50/50 seeds. A segmented-regression changepoint analysis likewise finds two log-loss segments that fit substantially better than one segment, but the third segment does not show the required post-plateau reacceleration. The median changepoint boundaries are epochs 40 and 255, with median slopes -0.1131 , -0.00229 , and -0.00036 for the three fitted segments. Thus our data support “rapid improvement followed by slow progress,” but not a robust three-stage sequence with a clearly detected exit.

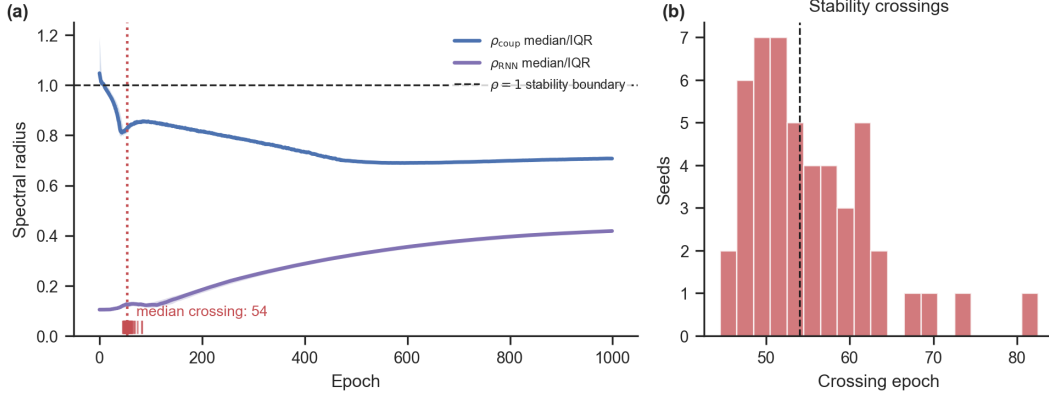


Figure 5: Coupled-system spectral analysis for the original setting. (a) Median coupled and RNN-only spectral radii across 50 seeds, with interquartile bands; the dashed line marks the $\rho = 1$ stability boundary and tick marks show per-seed crossing epochs. (b) Distribution of stability-crossing epochs. The coupled spectral radius crosses below the stability boundary early in training, whereas the RNN-only spectral radius remains far from the relevant stability threshold.

8 Result 4: Does the Phenomenon Generalize?

Generalization is hierarchical rather than uniform (Figure 6). Across 90 completed architecture/task extension runs, C1 deployed-loss divergence is supported in 42/90 runs. Architecture transfer within the double-integrator-style control family is strong: tanh RNN, GRU, and low-rank variants each show C1 support in 10/10 seeds. Task transfer to tracking is partial, with deployed-loss divergence in 8/10 seeds. Task-family transfer to ring/path-integration variants is weak: the simple ring variant has 0/10 supporting seeds and mean deployed-loss gap 5.4×10^{-4} .

The negative ring result is interpretable as a boundary condition rather than a direct contradiction of the original double-integrator claim. The simple ring task gives the learner $\sin(\text{error})$, $\cos(\text{error})$, and velocity, with a fixed target. This makes the deployed state-action mapping comparatively easy for the open-loop student to imitate. We therefore ran a harder partial-observation moving-target ring follow-up. It still shows only weak C1 support: 3/20 GRU seeds and 1/20 tanh seeds pass the deployed-loss criterion, with mean deployed-loss gaps 0.0837 and 0.0085, respectively. These results sharpen A2: the closed/open divergence transfers across architecture changes on the double-integrator-style control problem, but it does not automatically transfer to path-integration variants.

C3 diagnostic transfer is only tested where the coupled spectrum is implemented. It is supported in 10/10 tanh RNN runs and 10/10 low-rank runs, but not reported for GRU or ring runs because the current spectral code does not linearize gated nonlinear controllers. Local Jacobian spectra for gated nonlinear controllers are needed before claiming diagnostic transfer to GRUs. This is another reporting lesson: a generalization claim about stability mechanisms should distinguish performance transfer from diagnostic transfer.

9 Claim-Level Audit Matrix

Table 5 is the main summary of the paper. It separates original claims C1–C3 from audit questions A1–A2, and attaches each to evidence, robustness or generalization results, and an actionable lesson.

10 Actionable Lessons for Closed-Loop RNN Studies

The goal of the audit is to produce reusable reporting practices. We propose the following checklist for closed-loop RNN papers.

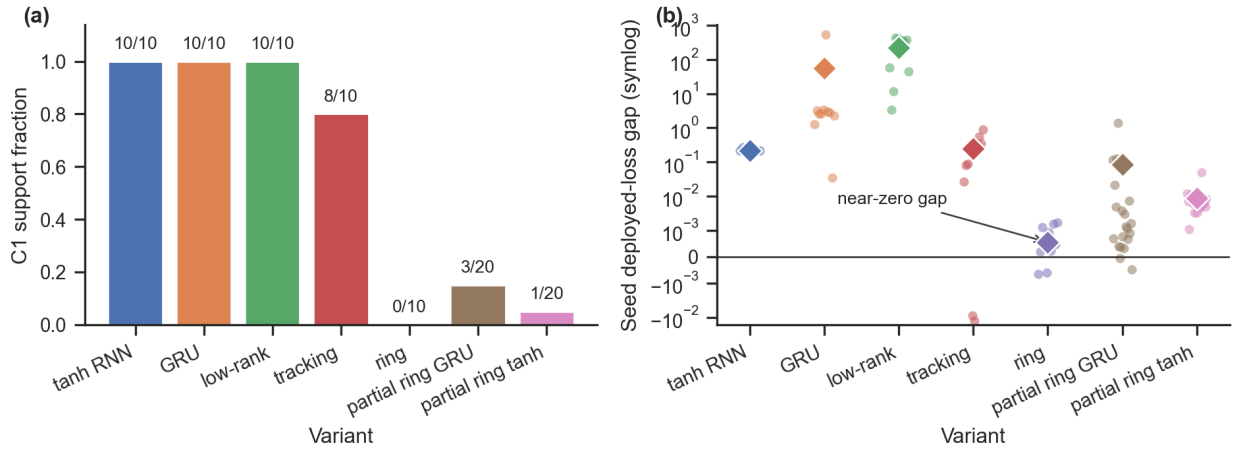


Figure 6: Generalization results across architecture and task variants. Standard variants use $n = 10$ paired seeds; the harder partial-observation ring variants use $n = 20$ paired seeds. (a) C1 support fractions with exact seed counts. (b) Seed-level deployed-loss gaps with mean diamonds on a symmetric-log scale, which permits tiny or negative gaps while still showing large architecture gaps. C1 transfers to tanh RNN, GRU, and low-rank variants, is partial on tracking, and is weak or absent in ring path-integration variants.

Table 5: Claim-level audit matrix for completed runs. C1–C3 are original claims under reproduction. A1–A2 are audit questions introduced here to test robustness, short/long horizon tradeoffs, and broader generalization.

Item	Evidence type	Original protocol	A1 audit	A2 audit	Lesson
C1: open/closed divergence	Paired seeds, effect size, CI, deployed loss	Reproduced: 50/50 seeds, loss gap 0.2186	Robust in most perturbations; absent under Adam and failed without clipping	42/90 extension runs; strong architecture transfer, weak path-integration transfer	Report deployed closed-loop loss, not only teacher-forced loss.
C2: stage structure	Stage detector, plateau duration, changepoints	Partial: slow progress phase 50/50, strict three-stage 0/50	Slow phase robust; strict exit boundary fragile	Slow phase common; strict boundary varies by task	Stage claims require algorithmic criteria, not visual identification.
C3: coupled stability	Coupled spectral radius crossing	Reproduced in linearizable setting: 50/50 seeds	Robust in all finite linearizable robustness settings	Holds for tanh/low-rank; not implemented for GRU/ring	Report coupled-system spectrum, not only RNN-only spectrum.
A1: protocol robustness and multi-horizon component tradeoff	Perturbation grid and component metric	N/A	Partial: union/radius 120/120, loss-only 0/120	N/A	Robustness claims need stress tests, explicit horizons, and component criteria.
A2: broader generalization	Architecture and task extensions	N/A	N/A	Partial: architecture transfer strong, tracking partial, ring weak	Closed-loop effects depend on task structure, environment coupling, and observability.

Report deployed closed-loop performance, not only open-loop or teacher-forced loss. The central risk is that a model can imitate a teacher under fixed inputs but fail when its own actions determine future inputs.

Use paired-seed comparisons when contrasting open-loop and closed-loop training. The clean comparison uses the same initialization, architecture, optimizer budget, and evaluation rollouts while changing only the training regime.

Define learning stages algorithmically. If a paper claims stages, it should specify transition criteria, minimum plateau duration, and how often each stage appears across seeds.

Analyze the coupled agent-environment system. RNN-only spectra can miss stability changes introduced by the environment. Coupled diagnostics should be reported whenever a mechanistic stability claim is made.

Report feedback strength and rollout horizon. These parameters determine how strongly actions affect future observations and can control whether the closed-loop phenomenon appears.

Report failure rates, not just average curves. Means can hide unstable runs, recovery failures, or seed-dependent plateaus.

Include at least one stress test. Optimizer, initialization, noise, feedback strength, and horizon perturbations are inexpensive compared with the interpretive cost of an untested mechanistic claim.

11 Discussion

This project treats Ger and Barak’s framework as a case study in reproducibility for interactive recurrent systems. The completed audit supports the two strongest original claims in the original setting: closed-loop and open-loop training diverge under deployment, and the coupled agent-environment system undergoes a stability transition in linearizable settings. The audit identifies three important limits. First, the stage claim depends strongly on how stages are defined. A slow-progress regime is reproducible, but a robust three-stage sequence with an objective second boundary is not recovered by our detectors. Second, A1 is only partially supported: the targeted sweep shows a stability-cost signal through coupled-radius increases, but not a loss-only short-vs-long horizon tradeoff. Third, generalization is hierarchical: architecture transfer within the double-integrator-style task is strong, tracking transfer is partial, and ring path-integration transfer is weak.

These mixed results are the reason a claim-level audit is useful. A figure-level reproduction would likely emphasize that the main closed-loop/open-loop loss gap appears. A methodological reproducibility study can say more: optimizer choice can remove the deployed-loss gap, missing gradient clipping can produce outright failure, path-integration-style variants can erase the gap even after adding partial observation and moving targets, spectra are only available for linearizable controllers unless local linearization is added, and stage claims should be accompanied by algorithmic boundaries rather than visual annotations alone. This is especially important for computational neuroscience, where RNNs are often used as mechanistic models and where closed-loop behavior is closer to biological learning than fixed-dataset supervision.

12 Limitations

The audit does not cover every RNN architecture, every control task, or full reinforcement-learning algorithms. The path-integration extensions are intentionally lightweight and should be interpreted as boundary-condition tests rather than a complete neuroscience benchmark. The A1 tradeoff metric is interval-level and diagnostic: it shows that short-horizon improvement often coincides with increased coupled spectral radius, but the loss-only horizon tradeoff is not supported in the current sweep. Finally, coupled spectral analysis is implemented for linearizable controllers and environments, but not yet for GRUs or other gated nonlinear controllers.

13 Artifact

The accompanying repository provides a config-driven implementation with smoke and full-run modes, raw CSV outputs, claim tables, and figure-generation scripts. The artifact is organized around the audit protocol rather than notebook-only reproduction: each experiment records the config, seed, device, metrics, and per-epoch time series. The public GitHub repository is archived through Software Heritage at https://archive.softwareheritage.org/browse/origin/?origin_url=https://github.com/aaravsinhaofficial/Auditing-Closed-Loop-Learning-in-Recurrent-Neural-Networks. Table 6 lists the main entry points.

Table 6: Artifact entry points. Full runtimes are approximate for one NVIDIA L40 GPU and are intended for planning rather than benchmarking.

Component	File/script	Produces	Runtime	Expected output
Smoke validation	<code>scripts/run_smoke.sh</code>	Tiny raw outputs, tables, figures	< 10 min	<code>results/raw/smoke_*</code>
Full audit	<code>scripts/run_full_audit.sh</code>	Core, robustness, generalization, figures	16–18 h	sweep summaries, figures
Targeted C2/A1/A2	<code>scripts/run_targeted_c2_a1_a2.sh</code>	Tradeoff, hard ring, changepoints	8–10 h	targeted summaries
Claim tables	<code>make_claim_tables</code> module	Claim matrix	< 1 min	<code>claim_matrix.csv</code>
Supplemental tables	<code>supplemental_audit_tables</code> module	Run accounting, C2 sensitivity, A1 split	< 1 min	processed CSVs
Figures	<code>make_all_figures</code> module	Paper figures	< 1 min	<code>figure_*.png</code>

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A Metric Definitions

Let s_t be the environment state, $o_t = Cs_t + \epsilon_t$ the observation, h_t the recurrent state, and $u_t = f_\theta(o_t, h_t)$ the action. Closed-loop deployment follows $s_{t+1} = F(s_t, u_t)$ and $h_{t+1} = G_\theta(o_t, h_t)$. The deployed loss is

$$L_{\text{deploy}}(\theta) = \frac{1}{T} \sum_{t=0}^{T-1} [\ell_s(s_t, s_t^*) + \lambda_u \|u_t\|_2^2], \quad (1)$$

while teacher-forced imitation loss compares the learner action to the teacher action on teacher-generated observations:

$$L_{\text{TF}}(\theta; \theta^*) = \frac{1}{T} \sum_{t=0}^{T-1} \|f_{\theta}(o_t^*, h_t) - f_{\theta^*}(o_t^*, h_t^*)\|_2^2. \quad (2)$$

For linearizable double-integrator settings we estimate an effective feedback gain by ridge regression, $\hat{K}_t = (X^\top X + \alpha I)^{-1} X^\top U$, and report the closed/open gain distance $D_K = (\sum_{t \in \mathcal{E}} \|\hat{K}_t^{\text{CL}} - \hat{K}_t^{\text{OL}}\|_F^2)^{1/2}$.

For linear RNN controllers and linear environments, with $s_{t+1} = As_t + Bu_t$, $u_t = W_{ho}h_t$, and $h_{t+1} = (1 - \eta)h_t + \eta(W_{hh}h_t + W_{ih}Cs_{t+1})$, the coupled transition matrix over $z_t = [s_t^\top, h_t^\top]^\top$ is

$$P_{\theta} = \begin{bmatrix} A & BW_{ho} \\ \eta W_{ih}CA & (1 - \eta)I + \eta W_{hh} + \eta W_{ih}CBW_{ho} \end{bmatrix}. \quad (3)$$

We report $\rho_{\text{RNN}} = \max_i |\lambda_i(W_{hh})|$ and $\rho_{\text{coup}} = \max_i |\lambda_i(P_{\theta})|$. The stability-crossing epoch is the first epoch at which $\rho_{\text{coup}} < 1$ for a minimum persistence window.

Stage labels use smoothed log deployed loss $y_t = \log(\max(L_{\text{deploy},t}, \epsilon))$. The detector looks for a fast initial negative slope, a sustained slow-progress interval, and a later reacceleration; the segmented variant fits three piecewise-linear segments and requires Stage 1 slope < -0.02 , Stage 2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage 3 slope $< \text{slope}_2 - 0.002$.

For A1, define a myopic-improvement interval by $\Delta \log L_e^{(T_s)} < -\delta_{\text{short}}$. We report four component criteria: union $I_s \wedge (I_\ell \vee I_\rho)$, loss $I_s \wedge I_\ell$, radius $I_s \wedge I_\rho$, and both $I_s \wedge I_\ell \wedge I_\rho$, where I_ℓ is $\Delta \log L_e^{(T_\ell)} > \delta_{\text{long}}$ and I_ρ is $\Delta \rho_{\text{coup},e} > 0$. In the targeted sweep, $T_s = 10$, $T_\ell = 200$, and $\delta_{\text{short}} = \delta_{\text{long}} = 0.005$.

For scalar paired metric M , the open/closed effect is $\Delta_i(M) = M_i^{\text{OL}} - M_i^{\text{CL}}$. We report mean, nonparametric bootstrap 95 percent CI, seed-level SD/IQR where useful, direction agreement $N^{-1} \sum_i \mathbf{1}[\Delta_i(M) > 0]$, and binary claim-support rate $S(c) = N^{-1} \sum_i q_i(c)$.